

Rummana Rahman

Email: rummanar@uci.edu

Ph: (925) 663-9785

LinkedIn: <https://www.linkedin.com/in/rummana-rahman11/>

Irvine, CA 92697, USA

SUMMARY

- Doctoral student at UC Irvine with expertise in embedded systems, digital design, machine learning, and IoT communications. Experienced in C, C++, and Python programming for hardware interfaces and simulations, with a strong foundation in microcontroller-based system development using platforms like Raspberry Pi and Arduino. Skilled in designing and simulating IoT systems and communication protocols and conducting extensive data analysis to inform system behavior. Adept in hardware and software integration with hands-on experience in embedded systems, time-series data analysis, hardware design tools like PSpice, Logisim, and MATLAB.
- 5+ years of teaching experience with undergrad courses in Computer Science majors and Electrical Engineering.
- 3+ years of industrial project management experience in Verizon Communications Inc. (New Jersey, USA) in Product Development, Network Deployment, Corporate Sustainability, and Content Delivery teams.

EDUCATION

Ph.D., University of California, Irvine, USA

Expected Date of Graduation Jun 2026

Department of Computer Science (CGPA 3.74/4)

Advisor: Prof. Nalini Venkatasubramanian

MSc., Rutgers, The State University of New Jersey, USA

Oct 2011

Electrical and Computer Engineering (CGPA 3.6/4)

BSc., University of Dhaka, Bangladesh

May 2008

Applied Physics, Electronics, and Communication Engineering, (1st class, equivalent to 4/4)

SKILLS

Hardware & Embedded Systems

- Microcontrollers: Raspberry Pi, Arduino, Node MCU
- Hardware Design & Simulation: PSpice, Multisim, Logisim, Simulink
- Interfaces: I2C, SPI, UART, GPIO
- Firmware: Embedded C, C++ for microcontroller programming
- Data Collection & Analysis: Sensor data (e.g., temperature, humidity), time-series data analysis
- Development Environments & OS: Linux, Raspbian, Arduino IDE, Jupyter Notebook, AWS, Colab, Anaconda

Programming & Tools

- Programming Languages: C, C++, Python, Bash, Assembly
- Version Control: Git, GitHub
- Simulation & Libraries: NS3, GNS3, MATLAB, TensorFlow, Keras, PyTorch, Pandas, NumPy
- Communication Protocols: LoRa, Wi-Fi, IoT, Networking

RELEVANT GRADUATE COURSES

Analog Transistor Circuit Design (A grade), Communication Networks I & II (A grade in I), Solid State Electronics, Optoelectronics I, Systems Engineering, Stochastic Signals and Systems, Modern Computer Systems and Architecture, Machine Learning, Systems and ML, Networking Lab, Advanced Network, Middleware Network and Distributed Systems

EXPERIENCE

Graduate Student Researcher

Oct 2021 to Present

Distributed Systems Middleware Group, Dept. of Computer Science, University of California, Irvine

- **Channel Modeling for Wireless IoT Network for Underground (IoUT) Communications:** Developed a physics-driven communication model for IoT sensors in underground environments (IoUT), including data collection, signal analysis, and simulation of wireless communication protocols using NS3 (C++).
- **Designed middleware** for time synchronization and data prediction using Python, C++, integrating distributed systems with IoT networks (for modeling network behavior, including delay, throughput, and packet drop rate).

Senior Lecturer

Dec 2015 to Sep 2021

Dept. of Electrical and Computer Engineering, North South University, Dhaka, Bangladesh

- **Courses Taught:** Digital Logic Design, Analog Electronics II, AC Circuits, Programming Language C. Junior Design Projects (involving Arduino, mobile apps and analog circuits). Updated lab manuals and lab equipment (oscilloscopes, signal generators, digital multimeters, PCs with latest version of simulators) during teaching tenure.

Rummana Rahman

Email: rummanar@uci.edu

Ph: (925) 663-9785

LinkedIn: <https://www.linkedin.com/in/rummana-rahman11/>

Irvine, CA 92697, USA

- **Course Projects:** Designed and debugged analog and digital circuits and heterogeneous sensors (using tools like PSpice, Multisim, and Logisim. Hardware tools Op-amps, filters, ADC/DAC, 555 timers, and i/o modules). Mentored students in Verilog and C++ programming for embedded systems and design verification.
- **Research projects with undergrad students:** Developed an **IoT monitoring system** using **Arduino, Raspberry Pi, and Node MCU** for low-income communities, for applications on sewage treatment plant, gas leakage detection, residential safety, and farmland monitoring (tools: Android Studio, Firebase, and other open source IoT platforms).

Engineer-III Cslt-Operations

Jul 2014 to Jan 2015

Network Deployment Team, New Product Development, Verizon Communications Inc., Basking Ridge, New Jersey

- Prepared Market Research Paper on Potential Datacenter: Researched and analyzed a multi-million dollar corporation's financial strength, market position, and product strength. data center and created a partner assessment document to assess revenue benefits through a potential alliance with that data center.
- Chaired touch-point calls with one of the top ten Data Centers in the USA and Verizon project managers to track the Verizon network deployment to the data center locations (forecasted revenue 30M\$ over 5 years)

Affiliate Network Engineer

Jul 2011 to Jul 2014

Engineering Leadership Development Program, Verizon Communications Inc., Basking Ridge, New Jersey

- **Affiliate Engineer-III, Corporate Sustainability Team:** Managed Verizon's Scope3 CO2 Reduction project and met the target of 200MT CO2 reduction for 2013. Worked with multinational tech corporates and services suppliers on toner recycling and printing reduction projects through which Verizon obtained volume reduction of paper prints (in millions per month) and saved Millions of BTUs of net energy.
- **Affiliate Engineer-II, Product Development, and Innovations:** Hosted and managed weekly multiple system operators (MSO) partner meetings with Time Warner Cable and Comcast. Maintained communication with MSO partners and cross-functional teams responding to the Verizon-MSO app integration issues and resolution support.
- **Affiliate Engineer, Content Delivery and Technologies:** Market research paper on MPEG DASH (Dynamic Adaptive Streaming over HTTP) standard for video streaming.

Graduated Research Assist.

Jun 2009 to Mar 2011

Advanced Photonics Lab, Dept. of Electrical and Computer Engineering, Rutgers, The State University of New Jersey

- **Optimizing surface texture of TCO for high efficiency solar cells:** Implemented texture-based light trapping technique within solar cell spectrum for thin film solar cell using ZnO as Transparent Conductive Oxide (TCO) with nano-scale textures. Obtained 21% conversion efficiency and 80% quantum efficiency on average (Tools: RSoft, Solar Cell utility).

SELECTED PUBLICATIONS

R. Rahman, C. H. Lin, C. H. Hsu and N. Venkatasubramanian, "A Measurement-informed Approach to Modeling Underground IoT Communications," *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*, Washington, DC, USA, 2024, pp. 1-7, doi: 10.1109/VTC2024-Fall63153.2024.10757716.

S. Rezwan, T. Ishtiak, R. Rahman, H. A. Rahman, M. Akter, H. A. Ratul, M. S. Hosain and M. Jakariya, "A Minimalist Model of IoT based Sensor System for Sewage Treatment Plant Monitoring," *2019 IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, BC, Canada, 2019, pp. 0939-0945, doi: 10.1109/IEMCON.2019.8936182.

MENTORING

IoT-SITY REU Graduate Student Mentor

Jul-Sep 2022, 2023, 2024

- Mentored REU batch of summer students on IoT research projects, involving hardware setups, including Raspberry Pi and sensor integration. with hands-on IoT projects involving raspberry pi, sensors, circuit building, and using Linux env.

ADDITIONAL INFORMATION

Languages: Fluent in English, Bengali

Tools: Familiarity with scripting languages such as Python and Bash, used for sensor data collection and network simulations

Firmware Experience: Hands-on experience with microcontroller programming using C/C++ for embedded systems development.